

# An Empirical Comparison of Human-Elicited and LLM-Generated Functional Requirements in a Physical Therapy Follow-Up System: A Paired-Analyst Study in Ecuador

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## Abstract

Digital support for home-based physical therapy depends on requirements that faithfully capture the needs of patients, caregivers, and physiotherapists. Large Language Models (LLMs) are increasingly proposed as assistants for requirements elicitation, but their comparative quality against human-elicited requirements remains underexplored in real-world, resource-constrained software engineering education settings. This paper reports a paired-analyst empirical study comparing the quality of functional requirements (FR) elicited by a human requirements engineering process against FR generated by an LLM (GPT-5.5-mini) from the same 18 anonymized interview transcripts, for the Intelligent System for Physical Therapy Control and Follow-up (SICST). The experimental comparison was registered on the Open Science Framework (OSF, DOI 10.17605/OSF.IO/82Q76) on 2026-09-12 (Ecuador time), after the field interviews had already been collected but before the documented LLM-generation and blind-evaluation phase. Therefore, the inferential comparison is reported as exploratory rather than as a fully preregistered confirmatory test. Thirty-three human-elicited and thirty-three LLM-generated FR (66 total) were anonymized and blind-rated by four independent judges across five quality dimensions (completeness, absence of ambiguity, verifiability, source correctness, internal consistency) on a 1–5 Likert scale. Inter-rater agreement was quantified with Fleiss' kappa. Eleven requirement pairs with strict thematic correspondence were compared using paired *t*-tests, with Holm–Bonferroni correction for multiple comparisons. Inter-rater agreement was

low to negligible across dimensions (Fleiss'  $\kappa$  ranging from  $-0.013$  to  $0.167$ ). No dimension showed a statistically significant difference between human- and LLM-generated requirements after correction (all adjusted  $p > 0.05$ ). These findings are interpreted as exploratory and inconclusive under the small paired sample, while highlighting the need for larger samples and more reliable evaluation protocols in future replications.

**Keywords:** Requirements elicitation, Physiotherapy informatics, Large Language Models, Empirical study, Patient tracking

## 1 Introduction

Physical therapy frequently extends beyond the clinical session. Patients may be expected to remember exercises, execute movements correctly, respect repetition and rest instructions, report pain or fatigue, and communicate problems between appointments. These activities create a requirements problem before they create an algorithmic problem: a digital system must first represent the needs, constraints, responsibilities, and safety expectations of patients, caregivers, and professionals in a way that can be verified and traced. Understanding whether Large Language Models can support this elicitation step, and at what quality cost, matters directly to small teams that lack dedicated requirements analysts.

Requirements engineering (RE) provides the methods needed to elicit, analyze, document, validate, prioritize, and manage stakeholder expectations (Pohl, 2010; ISO/IEC/IEEE, 2018; Sommerville, 2016). Recent work has begun to examine whether LLMs can support or replace parts of this process, reporting mixed results on completeness, ambiguity, and traceability when LLM output is compared against human analysts (Cheng et al, 2026). However, most of this evidence comes from generic or large-scale software domains; comparative studies in small, real-world, health-adjacent domains such as home-based physical therapy remain scarce, and few report a preregistered, blind, paired evaluation design.

SICST (*Sistema Inteligente de Control y Seguimiento de Terapia Física*) was developed as an academic case study in Requirements Engineering at Universidad Técnica Estatal de Quevedo. The project addresses follow-up of physical therapy routines by separating patient and physiotherapist interactions while maintaining a common traceability chain from field evidence to requirements, design elements, and prototype behavior. This paper reports the terminal empirical component of that project: a controlled comparison of requirements quality between the team's own elicitation process and an LLM prompted with the same underlying field evidence.

This study contributes: (1) a documented, blind, paired-analyst protocol for comparing human- and LLM-generated functional requirements in a real, non-simulated domain, with an OSF registration that preceded the documented LLM-generation and blind-evaluation phase but followed field elicitation; (2) an exploratory empirical estimate of the quality gap between both sources across five standard RE quality dimensions; and (3) a transparent report of inter-rater reliability challenges when four

independent, non-expert judges rate requirements quality on a Likert scale, which we discuss as a methodological caution for similar undergraduate empirical RE studies.

*RQ1.* Do quality dimensions (completeness, absence of ambiguity, verifiability, source correctness, internal consistency) differ between functional requirements elicited by a human requirements engineering process and those generated by a Large Language Model, when both are derived from the same anonymized interview transcriptions?

**H0.** There are no statistically significant differences in the quality of functional requirements generated through a human process versus those generated through a language model.

**H1.** There are statistically significant differences in the quality of functional requirements generated through a human process versus those generated through a language model.

## 2 Background and Related Work

### 2.1 LLMs for requirements elicitation

Generative language models have been proposed for several RE sub-tasks, including requirements extraction from unstructured text, ambiguity detection, and requirements quality assessment (Cheng et al, 2026). Evidence on elicitation quality specifically is more limited: most published comparisons use public or synthetic corpora rather than transcripts from a live field study conducted under documented informed-consent procedures, and few register the comparison protocol before generation, which leaves open the possibility of post-hoc rationalization of favorable results.

### 2.2 Requirements quality dimensions

Completeness, absence of ambiguity, verifiability, traceability to source, and internal consistency are long-standing quality attributes for individual requirements statements (ISO/IEC/IEEE, 2018; Sommerville, 2016). We adopt five of these dimensions, adapted to allow blind comparison between two requirement sources drawn from the same evidentiary base, rather than an absolute compliance checklist.

### 2.3 Physical therapy follow-up and telerehabilitation

Digital follow-up of physical therapy raises additional constraints beyond generic software requirements: exercise adherence, pain and fatigue self-reporting, caregiver involvement, and the professional’s authority to override any automated suggestion (ISO/IEC, 2023; Chazette and Schneider, 2020). These constraints motivated the original field study from which both requirement sets (human and LLM) were derived.

### 2.4 Inter-rater agreement in requirements evaluation

Fleiss’ kappa is a standard measure of agreement among three or more raters using categorical or ordinal scales (Fleiss, 1971; Cohen, 1960). Low agreement in subjective

quality ratings is a known risk in small-panel, non-expert evaluation designs and is reported transparently here rather than adjusted away.

## 3 Methodology

### 3.1 Design

This is a non-experimental, paired comparative study. No causal relationship between requirement source and quality is inferred; the study only compares the perceived quality of two requirement sets drawn from the same source evidence. The experimental comparison was registered on OSF (DOI [10.17605/OSF.IO/82Q76](https://doi.org/10.17605/OSF.IO/82Q76)) at 2026-09-13T00:50:38Z (2026-09-12 19:50:38 in continental Ecuador). The first documented field-elicitation sessions in the repository are dated 2026-06-17, so the OSF registration was not prior to the original interview corpus. The registration is therefore interpreted as governing the subsequent human-LLM comparison, including the documented LLM-generation and blind-evaluation phase, rather than as a preregistration of the earlier field collection.

### 3.2 Population and sampling

The unit of analysis is the individual functional requirement (FR). Two comparison groups were constructed from the same evidentiary source:

- **Human-elicited group** ( $n = 33$ ): FR produced through the team’s standard requirements engineering process (interviews, thematic analysis, and consolidation into the project’s ERS/SRS).
- **LLM-generated group** ( $n = 33$ ): FR produced by prompting GPT-5.5-mini with the same 18 anonymized interview transcripts used for human elicitation (physiotherapists, family caregivers/patients, and validation walkthrough participants), under a strict instruction set requiring each requirement to be grounded exclusively in the supplied transcripts (see Section 3.3).

Both sets were combined into a single list of 66 anonymized items and presented to four independent, blind judges (Section 3.4).

### 3.3 LLM requirements generation

The LLM (GPT-5.5-mini, accessed via ChatGPT) was given, in a single turn, the full text of the 18 anonymized interview transcripts and an explicit rule set requiring: (a) grounding every requirement strictly in the supplied transcripts, with no external knowledge or invented functionality; (b) a fixed sentence template ( “*The system shall [verifiable action] [object/condition]*”); (c) specificity, verifiability, and traceability to a concrete transcript fragment; (d) no duplicate or overlapping requirements; (e) sequential numbering (RF-01 to RF-33); and (f) output restricted to a CSV with columns `id_requisito`, `descripcion`, with no additional prose. A second turn asked the model to confirm its exact version and confirm that no external information was used. The full prompt, both turns, the model’s confirmation, and the resulting CSV are versioned in `06.Experimento/prompts_llm/` of the project repository.

### 3.4 Blinding and evaluation instrument

The 66 FR (33 human, 33 LLM) were normalized to a common format, shuffled, and assigned anonymous item identifiers with no indication of source. Four independent judges, blind to the origin of each item, rated every item on a 1–5 Likert scale across five dimensions: completeness, absence of ambiguity, verifiability, source correctness, and internal consistency. The worksheet delivered to evaluators contained only the anonymous ITEM identifiers and requirement text, with no source label. After all four evaluations had been completed, the mapping ITEM  $\rightarrow$  real requirement identifier  $\rightarrow$  source (human/LLM) was reconstructed from the versioned human traceability matrix and the LLM requirement set and exported as `06_Experimento/resultados/mapa_origen_items_reconstruido.csv` for reproducible analysis. This source mapping was not available to the evaluators during scoring.

### 3.5 Thematic pairing

To support a paired statistical comparison, human and LLM requirements addressing the same underlying interview theme were matched by two independent reviewers into a traceability matrix (`matriz trazabilidad tema RF.csv`). Eleven pairs achieved strict, unambiguous thematic correspondence (type `pareado`); these are the pairs used in the main exploratory paired analysis reported in Section 4. The remaining items (partial matches, weak matches, or items with no thematic counterpart in the other set) are retained in the traceability matrix for transparency but excluded from the paired test, since they do not support a valid 1-to-1 comparison.

### 3.6 Statistical analysis plan

For each of the five quality dimensions and for the aggregate score, the analysis (i) computed Fleiss’ kappa across all four raters and 66 items to quantify inter-rater agreement; (ii) tested normality of the 11 within-pair differences (human – LLM) with the Shapiro–Wilk test; (iii) applied a paired  $t$ -test when normality was not rejected, or a Wilcoxon signed-rank test otherwise; and (iv) corrected the five dimension-level  $p$ -values for multiple comparisons using the Holm–Bonferroni procedure. A 95% bootstrap confidence interval (10,000 resamples) was computed for the mean paired difference. All analyses were executed by a single versioned script (`06_Experimento/scripts analisis/analisis_experimento_llm_humano.py`) that reproduces every table and figure reported below directly from the raw blind-rating spreadsheets.

### 3.7 Deviations from the preregistered protocol

The exact statistical test per dimension (paired  $t$  vs. Wilcoxon) and the Holm–Bonferroni correction were specified operationally after registration, since the OSF protocol described the general comparison strategy without fixing the individual test names; this operational rule was fixed *before* the data were inspected for significance, and is documented in `06_Experimento/osf_deviations.pdf`. A second limitation

concerns timing: the field interview corpus used as source evidence already existed before the OSF registration. Accordingly, the registration is not presented as prior registration of the field data collection, and the statistical comparison is interpreted as exploratory with respect to that chronology.

## 4 Results

### 4.1 Inter-rater agreement

Agreement among the four blind raters was low to negligible across all five dimensions (Table 1). This indicates that the raters frequently disagreed on how to score individual requirements on the 1–5 scale, independently of requirement source.

**Table 1** Fleiss’ kappa by quality dimension (all 66 items, 4 raters).

| Dimension            | Fleiss’ $\kappa$ |
|----------------------|------------------|
| Completeness         | 0.132            |
| Absence of ambiguity | 0.167            |
| Verifiability        | −0.013           |
| Source correctness   | −0.011           |
| Internal consistency | 0.046            |

### 4.2 Supporting field-study descriptives and qualitative reliability

The reproducible field-data package under `07_Datos/resultados/` contains 79 questionnaire responses: 62 patients or former physical-therapy patients, 14 family caregivers, and 3 physiotherapists. Among the 76 responses to the camera-acceptance item, 43 (56.6%) selected *yes, but only with prior authorization* and 21 (27.6%) selected *yes*. Among the 76 responses to the progress-visualization item, 64 (84.2%) answered *yes*.

The independent qualitative double-coding check compared 36 coding decisions, with 33 agreements and 3 disagreements (91.67% raw agreement). Cohen’s  $\kappa$  was 0.7187 with a 95% bootstrap interval of [0.3077, 1.0000], based on 10,000 resamples with seed 401. This statistic describes agreement between the two qualitative coders and is distinct from the Fleiss’  $\kappa$  values above, which describe agreement among the four blind requirement raters.

The thematic-saturation analysis accumulated 17 themes across the 18 canonical elicitation sessions. The last new theme appeared at session 9 (EFT-01); sessions 10–18 contributed no additional themes. These results are generated by the reproducible analysis chain and are reported here to connect the empirical field evidence with the requirement-comparison study.

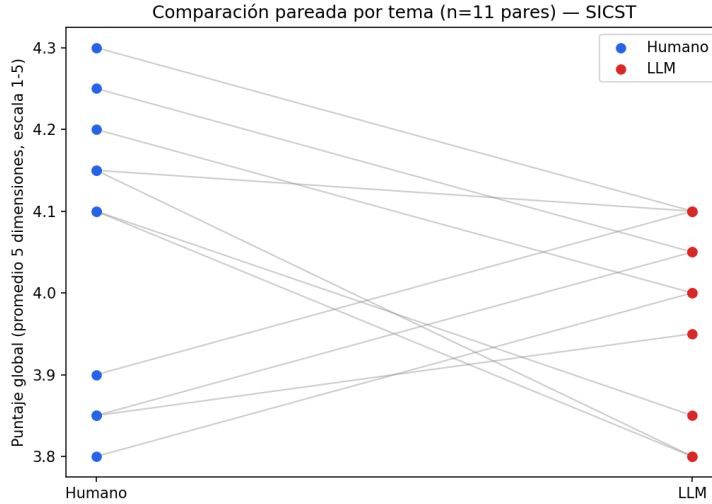
### 4.3 Paired comparison (RQ1)

Table 2 reports the paired analysis over the 11 thematically-matched requirement pairs. All five within-pair difference distributions did not reject normality (Shapiro–Wilk  $p > 0.05$ ), so a paired  $t$ -test was used throughout. After Holm–Bonferroni correction for the five dimension-level comparisons, no dimension reached statistical significance.

**Table 2** Paired comparison, human-elicited vs. LLM-generated FR ( $n = 11$  pairs). Mean difference is human minus LLM.

| Dimension            | $t(10)$ | Mean diff. | 95% CI            | $p$ (raw) | $p$ (Holm) |
|----------------------|---------|------------|-------------------|-----------|------------|
| Completeness         | 0.714   | 0.091      | $[-0.136, 0.341]$ | 0.492     | 1.000      |
| Absence of ambiguity | -0.468  | -0.091     | $[-0.455, 0.250]$ | 0.650     | 1.000      |
| Verifiability        | 1.000   | 0.136      | $[-0.114, 0.386]$ | 0.341     | 1.000      |
| Source correctness   | 1.550   | 0.159      | $[-0.045, 0.341]$ | 0.152     | 0.761      |
| Internal consistency | 0.938   | 0.091      | $[-0.091, 0.273]$ | 0.371     | 1.000      |

Figure 1 shows the paired global scores (average of the five dimensions) for each of the 11 matched themes, illustrating the substantial overlap between the two sources.



**Fig. 1** Paired global quality score by thematic pair, human-elicited vs. LLM-generated requirements ( $n = 11$ ).

We cannot reject  $H_0$  for any of the five quality dimensions. The complementary global paired score also did not show a statistically significant difference (mean human-minus-LLM difference 0.0773, 95% bootstrap CI  $[-0.0455, 0.1955]$ ,  $t(10) = 1.193$ ,  $p = 0.261$ ). Within the limits of this small-sample exploratory design, the observed

ratings do not provide evidence of a difference in perceived quality between the two sources; this should not be interpreted as evidence of equivalence.

## 5 Discussion

### 5.1 Implications for research

The absence of a detectable quality gap, combined with very low inter-rater agreement, is itself an informative result: it suggests that, for short, templated functional requirements derived from the same evidentiary transcripts, non-expert raters struggle to reliably distinguish machine-generated from human-generated statements. This is consistent with the broader concern that LLM output can closely mimic the surface form of expert-authored technical text (Cheng et al, 2026), which may complicate quality assurance processes that rely on manual review.

### 5.2 Implications for practice

For small teams without dedicated requirements specialists, an LLM prompted directly and strictly from anonymized field transcripts produced requirements that blind raters did not judge as worse, on average, than the team’s own elicitation process. This does not license a general recommendation to replace human elicitation; it does suggest that LLM-assisted drafting, followed by human review and thematic validation, is a defensible starting point for resource-constrained projects.

### 5.3 A surprising or counterintuitive finding

The negative and near-zero Fleiss’ kappa values for verifiability and source correctness are notable: they indicate that, on those two dimensions, the four raters agreed less than chance would predict. This is unusual and suggests the rating instrument itself, not only the requirements, deserves revision in future replications (e.g., anchored behavioral examples per Likert point, or a smaller, better-trained rater panel).

We restrict all conclusions to this specific, small-sample, Ecuadorian, physical-therapy domain study; no claim is made that LLMs match human analysts in general software domains, sample sizes, or languages.

## 6 Threats to Validity

Following common software-engineering case-study practice (Runeson and Höst, 2009; Wohlin et al, 2012), threats are organized into construct, internal, external, and conclusion validity.

### 6.1 Construct validity

The five quality dimensions operationalize a subjective judgment on a 1–5 scale without behaviorally anchored rating examples, which may partly explain the low inter-rater agreement observed. The main exploratory paired test uses only the 11

strictly-matched pairs; broader, weaker, or unmatched items (44 of 66) are not part of the paired statistical claim.

## 6.2 Internal validity

Raters were blind to requirement origin, which mitigates expectation bias, but they were not trained requirements engineers, which may explain the low kappa values. The Holm–Bonferroni correction and the choice between paired  $t$  and Wilcoxon tests were fixed by a rule specified before inspecting results (Section 4 deviations), reducing the risk of post-hoc test selection. However, the original field interviews predated the OSF registration and the operational statistical rule was specified after registration; for these reasons, the inferential comparison is reported as exploratory rather than as a fully preregistered confirmatory analysis.

## 6.3 External validity

The work concerns one academic project, one LLM (GPT-5.5-mini), one prompting strategy, and a physical-therapy domain in a specific Ecuadorian context. Results should not be generalized to other LLMs, prompting strategies, domains, or rater populations without further replication.

## 6.4 Conclusion validity

With only 11 paired observations, statistical power to detect small-to-moderate effect sizes is limited; the failure to reject  $H_0$  should not be interpreted as strong evidence of true equivalence, but as an inconclusive result under low power, reported transparently rather than reframed as a positive finding.

# 7 Ethics, Privacy, and Data Governance

The underlying interview transcripts used to generate both requirement sets were collected under the project’s documented informed-consent procedures and are stored in anonymized form in the public research materials, while direct identifiers and signed consent evidence are restricted to an encrypted, access-controlled container. The blind requirement-evaluation worksheet contained only anonymous ITEM identifiers and requirement text. The source mapping used for analysis was reconstructed only after all four evaluations had been completed and is published as a reproducibility artifact; it was not available to evaluators during scoring.

# 8 Reproducibility and Open Materials

The complete experimental package is versioned in the project repository under `06_Experimento/`: the registered protocol (`protocolo.pdf`), the OSF registration receipt (`osf_registration.pdf`), the deviations declaration (`osf_deviations.pdf`), the exact LLM prompts and confirmations (`prompts_llm/`), the raw blind-rating spreadsheets and the LLM-generated requirements (`datos_crudos/`), the thematic pairing matrix (`datos_procesados/`), the single reproducible analysis

script (`scripts analisis/`), and every table and figure reported in this paper (`resultados/`). Running the analysis script against the raw data reproduces Tables 1–2 and Figure 1 exactly.

Dataset DOI: <https://doi.org/10.5281/zenodo.22315298> | OSF registration: <https://doi.org/10.17605/OSF.IO/82Q76>

## 9 Conclusion

This paper reported an OSF-registered, blind, paired-analyst comparison of the quality of functional requirements elicited by a human process versus generated by an LLM (GPT-5.5-mini), for a physical therapy follow-up system, using the same 18 anonymized interview transcripts as the evidentiary source for both. The field interviews predated the OSF registration, while the documented LLM-generation and blind-evaluation phase followed it; accordingly, the statistical comparison is interpreted as exploratory. Inter-rater agreement among four blind judges was low to negligible across all five evaluated quality dimensions. After correcting for multiple comparisons, no dimension showed a statistically significant difference between human-elicited and LLM-generated requirements (all adjusted  $p > 0.05$ ); the null hypothesis could not be rejected.

These results, obtained under a small paired sample ( $n = 11$ ), should be read as a transparent, inconclusive finding rather than evidence that LLMs match human requirements engineers in general. Future work should increase the number of thematically-matched pairs, train and calibrate the rater panel to improve inter-rater agreement, and replicate the comparison across additional LLMs and domains before drawing practice-level conclusions.

## Declarations

**Funding.** This academic project reports no external research funding in the materials used for this manuscript.

**Conflict of interest.** The authors declare no conflict of interest relevant to this manuscript.

**Author contributions.** Angelo Paul Zambrano Moya led the OSF registration, LLM requirements generation, blind-evaluation coordination, and statistical analysis. Kevin Germán Contreras Chávez contributed as a co-interviewer in field data collection sessions conducted remotely. Both authors reviewed the manuscript and are responsible for verifying that no evidence reported here was fabricated or altered.

**Data availability.** The reproducible dataset is available at Zenodo, DOI: <https://doi.org/10.5281/zenodo.22315298>. The registered experimental protocol is available at OSF, DOI: <https://doi.org/10.17605/OSF.IO/82Q76>. Public materials exclude direct identifiers and signed consent documents.

**Code availability.** The project source repository, including the analysis script that reproduces every reported table and figure, is publicly available at [https://github.com/gleiston-guerrero/SICST\\_ISR401](https://github.com/gleiston-guerrero/SICST_ISR401).

**Use of AI tools.** An LLM (GPT-5.5-mini) was used to generate one of the two compared requirement sets, as the object of study itself; this use is fully documented in `06.Experimento/prompts_llm/`. Separately, AI assistance was used for repository organization, technical writing review, data-processing script development, and manuscript drafting from author-provided real data and results, as documented in `10.Autoria/declaracion_uso_ia.md`. No participant, response, interview, questionnaire, or empirical result was invented or altered by this assistance.

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